Solution Set

Question 1

Let X and Y be risk-factor shocks with joint density f(x,y). Derive the marginal density of X and the conditional density of Y given X=x.

Answer key

The marginal density integrates out Y:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy$$

If f_X(x)>0, the conditional density is

$$f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$$

The marginal asks what X does alone; the conditional asks how Y behaves after observing X=x.

Question 2

For jointly normal

$$X, Y$$

with means 0, volatilities

$$\sigma_X, \sigma_Y$$

and correlation

$$\rho$$

state

$$\mathbb{E}[Y \mid X = x]$$

and

$$\mathrm{Var}(Y \mid X = x)$$

Answer key

The conditional distribution is normal with

$$\mathbb{E}[Y \mid X = x] = \rho \frac{\sigma_Y}{\sigma_X} x$$

$$\mathrm{Var}(Y \mid X = x) = \sigma_Y^2 (1 - \rho^2)$$

This is the analytical basis for regression-style stress propagation in Gaussian factor models.

Question 3

A portfolio loss is

$$L = 3X + 5Y$$

where X and Y have volatilities 2 and 4 and correlation 0.6. Compute

$$\mathrm{Var}(L)$$

Answer key

The covariance is

$$\mathrm{Cov}(X, Y) = 0.6(2)(4) = 4.8$$

Therefore

$$\mathrm{Var}(L) = 9(4) + 25(16) + 2(3)(5)(4.8) = 36 + 400 + 144 = 580$$

Question 4

Explain tail dependence and why ordinary correlation can miss it.

Answer key

Tail dependence measures the tendency of variables to experience extreme losses together. Linear correlation summarizes average co-movement around the center of the distribution. Two models can have the same correlation but very different joint crash probabilities, which is why copulas and stress scenarios matter.

Question 5

Two default indicators have P(D1=1)=0.04, P(D2=1)=0.06, and P(D1=D2=1)=0.01. Test independence and interpret.

Answer key

Independence would require

$$\mathbb{P}(D_1 = 1, D_2 = 1) = 0.04(0.06) = 0.0024$$

The observed joint default probability is 0.01, much higher. Defaults are positively dependent. In credit portfolios this greatly increases tail loss.

Question 6

Derive the covariance matrix of a two-factor linear PnL vector

$$\Delta\Pi = BX$$

where

$$X$$

has covariance

$$\Sigma_X$$

Answer key

The covariance matrix is

$$\mathrm{Cov}(\Delta\Pi) = \mathrm{Cov}(BX) = B\mathrm{Cov}(X)B^\top = B\Sigma_X B^\top$$

This formula is central to factor-model VaR: exposures map factor covariance into portfolio PnL covariance.

Question 7

If X,Y are independent continuous variables, prove f_{X,Y}(x,y)=f_X(x)f_Y(y). What is the risk interpretation?

Answer key

For independent variables, joint probabilities over rectangles factor:

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B)$$

For continuous variables this implies the joint density factorization

$$f_{X,Y}(x,y) = f_X(x) f_Y(y)$$

Risk interpretation: knowing one factor shock gives no distributional information about the other.

Question 8

A Gaussian copula uses normal latent variables to join non-normal marginals. Explain the construction in three steps.

Answer key

First simulate correlated standard normals Z. Second transform them to uniforms using U_i = Phi(Z_i). Third transform each uniform through the desired marginal inverse CDF:

$$X_i = F_i^{-1}(U_i)$$

This separates marginal distribution choice from dependence modeling, although Gaussian copulas may understate joint tail dependence.

Question 9

For bivariate normal shocks, why is a linear portfolio PnL exactly normal but an option PnL not exactly normal?

Answer key

A linear portfolio PnL is a linear combination of jointly normal variables, so it is normal. An option PnL contains nonlinear terms such as gamma:

$$\Delta V \approx \Delta S + \tfrac{1}{2}\Gamma(\Delta S)^2$$

The squared normal term is not normal, so the option PnL distribution is generally skewed or curved even when the underlying shock is normal.

Question 10

Define covariance matrix positive semidefiniteness and link it to joint distributions.

Answer key

A covariance matrix Sigma is positive semidefinite when

$$a^\top \Sigma a \geq 0 \quad \text{for every vector } a$$

This must hold because a^T Sigma a is the variance of the linear combination a^T X. Any proposed joint distribution with a non-PSD covariance matrix is invalid.

Question 11

Explain why diversification benefit is nonlinear in correlation near high correlations.

Answer key

Portfolio variance includes the term

$$2w_1 w_2 \rho \sigma_1 \sigma_2$$

As rho approaches 1, the portfolio behaves more like one concentrated bet. The remaining diversification benefit shrinks, and in stress settings correlations often rise exactly when diversification is needed most.

Question 12

A joint model fits marginal VaR for each desk but underestimates portfolio VaR. What model weakness does that suggest?

Answer key

The marginals may be acceptable while dependence is wrong. Underestimated portfolio VaR often suggests understated correlations, missing tail dependence, stale data alignment, or failure to capture common stress drivers. Joint modeling is not validated by marginal backtests alone.